

TECHNICAL WHITEPAPER // NDL V4.1

The Elimination of Instructional Jitter and Thermal-Drift via Bilateral Determinism

Status: Proposed Standard

Revision: 4.1.0-A

Subject: Hardware-Agnostic Mathematical Sovereignty

1. ABSTRACT

Modern embedded systems rely on a "Good Enough" model of determinism, where software timing is assumed but rarely verified at the bit-level across disparate silicon architectures. The Nexorian Deterministic License (NDL V4.1) introduces a framework for Mathematical Sovereignty, ensuring that logic execution remains bit-identical regardless of physical hardware (ARM, x86, RISC-V) or environmental thermal drift.

2. THE PROBLEM VECTOR

Traditional systems face three primary "Noise" vectors that prevent true autonomy in high-liability environments such as gas pipeline fail-safes and motor control loops:

- **IEEE 754 Variance:** Floating-point rounding errors that differ between FPU implementations.
- **Thermal Drift:** Voltage fluctuations causing minor timing variations in high-frequency oscillators.
- **Instructional Jitter:** Non-deterministic branching and cache-miss latencies inherent in modern out-of-order execution CPUs.

3. THE MANDATES

3.1 Integer-Only State Machines

To achieve bit-identical results, NDL V4.1 forbids the use of standard floating-point arithmetic in safety-critical kernels. All calculations must utilize Q-format fixed-point math to eliminate the 1-bit rounding "drift" that causes system desync over billions of cycles.

3.2 Deterministic Memory Mapping (DMM)

Safety-critical partitions must bypass non-deterministic OS features like ASLR (Address Space Layout Randomization). By enforcing static, deterministic memory addresses, we ensure "Time-to-Compute" remains a constant, not a variable.

3.3 Bilateral Hash Verification

The NDL V4.1 requires that a specific instruction set must produce a bit-identical output hash across disparate architectures. Whether running on a \$10 MCU or a server-grade processor, the resulting system-state hash must match perfectly.

4. CONCLUSION

The NDL V4.1 is a declaration that Determinism is the precursor to Autonomy. We no longer "trust" hardware to be consistent; we force it to be bit-identical. This whitepaper serves as the foundational logic for the MB-5D prediction engine and Nexorian sovereign infrastructure.

"Proof before Trust, Rules before Reasoning, & Determinism before Autonomy."